

$$3d^8$$

September 3, 2026

$$1 \quad \left| 3d_{yz}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 3d_{xz}^\alpha 3d_{xy}^\alpha \right\rangle$$

Here are the CSFs:

$$|1, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|2, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|3, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$|4, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|5, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$|6, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle$$

$$|7, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|8, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$|9, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle$$

$$|10, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle$$

$$|11, 1, 0\rangle = \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|12, 1, 0\rangle = \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$\begin{aligned}
|13, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|14, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|15, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|16, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|17, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|18, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|19, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|20, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|21, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|22, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|23, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|24, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|25, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|26, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|27, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|28, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|29, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|30, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|31, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|32, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|33, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|34, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|35, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|36, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|37, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|38, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|39, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|40, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|41, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|42, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|43, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|44, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|45, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

The Ground State CSFs are 9 19 29

$$\begin{aligned}
\langle 9, 1, 1 | L_y | 2, 1, 1 \rangle &= I \\
\langle 9, 1, 1 | L_y | 3, 1, 1 \rangle &= -\sqrt{3}I \\
\langle 9, 1, 1 | L_y | 10, 1, 1 \rangle &= I
\end{aligned}$$

$$\Delta g_{yy}$$

$$\begin{aligned}
& +\zeta\Delta_{9-2}^{-1} \\
& +3\zeta\Delta_{9-3}^{-1} \\
& +\zeta\Delta_{9-10}^{-1}
\end{aligned}$$

$$\begin{aligned}
\langle 9, 1, 1 | L_z | 6, 1, 1 \rangle &= -I \\
\langle 9, 1, 1 | L_z | 7, 1, 1 \rangle &= 2I
\end{aligned}$$

$$\Delta g_{zz}$$

$$\begin{aligned}
& +\zeta\Delta_{9-6}^{-1} \\
& +4\zeta\Delta_{9-7}^{-1}
\end{aligned}$$

For a multiplicity of 3 we have the following:

$$D = \langle 1, 1 | 1, 1 \rangle - \langle 1, 0 | 1, 0 \rangle$$

$$E = \langle 1, -1 | 1, 1 \rangle$$

$$\begin{aligned}
\left\langle 1, 1 \left| \hat{H}_{eff} \right| 1, 1 \right\rangle &= -\zeta^2 (\\
& + \frac{1}{4}\Delta_6^{-1} + \Delta_7^{-1} + \frac{1}{8}\Delta_{12}^{-1} + \frac{3}{8}\Delta_{13}^{-1} + \frac{1}{8}\Delta_{20}^{-1} + \frac{1}{8}\Delta_{32}^{-1} \\
& + \frac{3}{8}\Delta_{33}^{-1} + \frac{1}{8}\Delta_{40}^{-1} + \frac{1}{4}\Delta_{43}^{-1} + \frac{1}{4}\Delta_{45}^{-1})
\end{aligned}$$

$$\begin{aligned}
\left\langle 1, 0 \left| \hat{H}_{eff} \right| 1, 0 \right\rangle &= -\zeta^2 (\\
& + \frac{1}{8}\Delta_2^{-1} + \frac{3}{8}\Delta_3^{-1} + \frac{1}{8}\Delta_{10}^{-1} + \frac{1}{8}\Delta_{22}^{-1} + \frac{3}{8}\Delta_{23}^{-1} + \frac{1}{8}\Delta_{30}^{-1} \\
& + \frac{1}{4}\Delta_{36}^{-1} + \Delta_{37}^{-1})
\end{aligned}$$

$$\begin{aligned}
\left\langle 1, -1 \left| \hat{H}_{eff} \right| 1, 1 \right\rangle &= -\zeta^2 (\\
& - \frac{1}{8}\Delta_{12}^{-1} - \frac{3}{8}\Delta_{13}^{-1} - \frac{1}{8}\Delta_{20}^{-1} + \frac{1}{8}\Delta_{32}^{-1} + \frac{3}{8}\Delta_{33}^{-1} \\
& + \frac{1}{8}\Delta_{40}^{-1} - \frac{1}{4}\Delta_{43}^{-1} - \frac{1}{4}\Delta_{45}^{-1})
\end{aligned}$$

$$\begin{aligned}
D = & \zeta^2 (\\
& + \frac{1}{8} \Delta_2^{-1} + \frac{3}{8} \Delta_3^{-1} - \frac{1}{4} \Delta_6^{-1} - \Delta_7^{-1} + \frac{1}{8} \Delta_{10}^{-1} - \frac{1}{8} \Delta_{12}^{-1} \\
& - \frac{3}{8} \Delta_{13}^{-1} - \frac{1}{8} \Delta_{20}^{-1} + \frac{1}{8} \Delta_{22}^{-1} + \frac{3}{8} \Delta_{23}^{-1} + \frac{1}{8} \Delta_{30}^{-1} - \frac{1}{8} \Delta_{32}^{-1} \\
& - \frac{3}{8} \Delta_{33}^{-1} + \frac{1}{4} \Delta_{36}^{-1} + \Delta_{37}^{-1} - \frac{1}{8} \Delta_{40}^{-1} - \frac{1}{4} \Delta_{43}^{-1} - \frac{1}{4} \Delta_{45}^{-1} \\
&)
\end{aligned}$$

$$\begin{aligned}
E = & \zeta^2 (\\
& + \frac{1}{8} \Delta_{12}^{-1} + \frac{3}{8} \Delta_{13}^{-1} + \frac{1}{8} \Delta_{20}^{-1} - \frac{1}{8} \Delta_{32}^{-1} - \frac{3}{8} \Delta_{33}^{-1} \\
& - \frac{1}{8} \Delta_{40}^{-1} + \frac{1}{4} \Delta_{43}^{-1} + \frac{1}{4} \Delta_{45}^{-1})
\end{aligned}$$